

Qi Xin

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IEEE Xplore Page

Google Scholar Page

Education

Queen Mary University of London	Jun. 2026 – Present
• Postgraduate Research Associate in Computer Science	
Beijing University of Posts and Telecommunications	Sep. 2022 – Present
• Ph.D. Candidate in Information and Communication Engineering	
Beijing Jiaotong University	Sep. 2018 – Jun. 2022
• B.S. in Communication Engineering; GPA: 3.82/4.00	

Research Interest

- AI networking for LLMs (Congestion control, network monitoring, and collective communication optimization)
- LLM framework optimization
- Geo-distributed LLM systems

Publications

[1] Megatuner: An Offline DCQCN Parameter Tuner for Large-Scale Model	2024
Qi Xin, Xiaoxiang Wang, He Liu, Fangzheng Jiao, Xiaohe Hu	
<i>Proc. IEEE Global Communications Conference (GLOBECOM)</i> , 2024.	
[2] MegaMon: Transformer-Based Network Traffic Monitor and Analysis for LLM Training	2025
Qi Xin, Xiaoxiang Wang, Dongyue Zhang, Bowen Han, Bohua Xu	
<i>Proc. IEEE International Conference on Communications (ICC)</i> , 2025.	
[3] VENUSSIM: Torch-level Execution Trace Simulator for LLM Training	Preprint
Qi Xin et al.	
[4] Activating the Power of Congestion Control for Large-Scale Model Training	Preprint
Qi Xin, Xiaoxiang Wang, Ahmed M. Abdelmoniem	
[5] When Do Low-Cost GPU Supernodes Help? Measurement, Topology-Communication Fit, and Cost-Performance Topology Search	Preprint
Qi Xin, Xiaoxiang Wang, Ahmed M. Abdelmoniem	
[6] CommPilot: Orchestrating Communication Flows for LLM training in Datacenter	Preprint
Qingkai Meng, Qi Xin, Chi Wang, Tian Chen, et al.	
[7] Cross-Data Center Training with Heterogeneous Accelerators: Protocols and Evaluation	2026
Bohua Xu, Qi Xin, Xiong-yan Tang, et al.	
<i>Proc. IEEE International Conference on Communications (ICC)</i> , 2026.	

Industry Experience

Infrawaves	AI Infrastructure Engineer (Intern)	Jun. 2023 - Jun. 2026
•	Stood up end-to-end LLM training clusters from scratch (10+ clusters, >30,000 GPUs across H100/H200, MetaX C500, PPU), completing network topology design, base images/drivers, and Docker/K8s orchestration to production readiness.	
•	Built a trace-driven, full-stack LLM training simulator (VenusSim) from torch-level execution traces.	
•	Developed Megatuner , an offline search framework for DCQCN parameters that identifies scale-appropriate configurations on VenusSim .	
•	Proposed MegaMon , a non-intrusive training-traffic monitor using passive collection and time-series modeling to localize QP-rate anomalies and link degradation at millisecond granularity without perturbing running jobs.	
•	Conducted cross-data-center training studies over 20/30/50/1000 km paths; refined pipeline-parallel scheduling and reported efficiency $\geq 97\%$ relative to local baselines under identical workloads.	
•	Designed an LLM-oriented congestion-control policy (shortest-remaining-packet-first with micro-batch-aware prioritization) and implemented it on MetaConnect-400S RDMA NICs, demonstrating effectiveness on long-haul and intra-DC traces.	
•	Contributed to the VCCL collective-communication library, including interface specification, operator implementations, and performance pathways for large-scale training.	

Skills

Skills: Kubernetes, Docker, Ray; Python, C/C++; PyTorch, Megatron-LM; Linux; NS-3, Prometheus, Grafana.